

Chapter 11

Laryngeal contrast in the Celtic languages: Variation, typology, and history

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This chapter addresses the nature of laryngeal contrast throughout the history of the Celtic languages. In the present day, there is significant phonetic and phonological variation across the languages in distinctions among two series of stops, while the historical data is ambiguous in multiple ways. The chapter re-evaluates the issues in light of some recent findings in phonetic and diachronic typology. I argue that the “aspirating” system of many present-day languages (contrasting aspirated /p^h t^h k^h/ stops with a variably voiced /b d g/ series) most likely arose around the mid 1st millennium CE “syncope period” in both Gaelic and Brythonic. I show there are significant difficulties in projecting it towards the earlier period, where a “voicing-based” system (unaspirated /p t k/ vs. categorically voiced /b d g/) is more compatible with the observed sound changes. This analysis implies that the rise of the aspirating system aligned with a suite of drastic phonological changes affecting the Insular Celtic languages in the sub-Roman period, and underscores the importance of synchronic variation for reconstructing the early history of the Celtic laryngeal contrast.

1 Introduction

The aim of this chapter is to consider the history of laryngeal contrast in the Celtic languages, especially the phonetics and phonology of stops. Most present-day Celtic languages distinguish between two series of stops, spelled in most



contexts as ⟨p t c/k⟩ and ⟨b d g⟩.¹ Their phonetic realisations vary non-trivially across the family: to avoid ambiguity, I will use the terms FORTIS and LENIS respectively to refer to the two series, eschewing the terminology of “voiced”, “voiceless”, or “(un)aspirated” to avoid confusion between phonological and phonetic description.² Synchronically, the present-day Celtic languages are fairly similar in the (morpho)phonological patterning of stops, but are considerably more diverse in how the contrast is phonetically implemented. Diachronically, the trajectories of sound change that led to the present-day states of the languages differ more significantly, and, as I will show in this chapter, this provides us with a wealth of information for phonetic analysis and typological comparison.

In order to consider the history of laryngeal contrast, in Section 2 I sketch the patterns observed in the present-day languages. Section 3 lays out the arguments made in favour of a view of earlier stages of Celtic as an “aspirating” language characterized by aspiration of the FORTIS series and variable voicing of the LENIS series of stops; in various guises, this analysis has long been commonly accepted as correct. In Section 4 I lay out the relevant sound changes and consider whether they are phonetically and typologically plausible from an “aspirating” starting point. Section 5 synthesizes the findings and lays out the key argument of this chapter that “aspirating” systems are a relatively recent innovation in the history of the attested Insular Celtic languages, possibly dating to no earlier than the mid 1st millennium CE. Section 6 concludes the chapter with a range of further considerations, with specific attention to reconstructing variation in the proto-language and its consequences particularly in the case of Celtic.

¹I use ⟨angled brackets⟩ to refer to orthographic representations, ALL CAPS for abstract phonological categories, [pipes] to signal phonological feature specifications, /slashes/ for phonological underlying specifications, and [square brackets] for phonological surface representations.

²The terms “fortis” and “lenis” themselves, of course, do also have phonetic connotations and have been used to refer to aspects of phonetic realisations: for a review, see Honeybone (2008). It would have been preferable to use even less suggestive terminology: one possibility is calling the ⟨p t k⟩ series *tenues* and ⟨b d g⟩ *mediae*, a usage formerly common especially in Indo-European (including Celtic) comparative philology. However, this terminology is now so unfamiliar that I have opted for “fortis” and “lenis”.

2 Laryngeal phonology in present day Celtic

2.1 Theoretical preliminaries

The analysis of laryngeal stop contrasts in this chapter is informed by the influential framework of Laryngeal Realism (a term due to Honeybone 2005). It was formulated as an alternative to a view common in mainstream generative phonology, in which two-term systems contrasting ⟨p t k⟩ and ⟨b d g⟩ series were generally analysed as specified [–voice] and [+voice], respectively. Under this “broad interpretation of the feature [±voice]” (Hall 2001), many cross-linguistic differences in the phonetic correlates of the two specifications are considered irrelevant, as long as voicing participates in the realisation of the contrast in some way. Therefore, languages like French [stan1290] and Ukrainian [ukra1253], in which FORTIS stops lack significant aspiration and LENIS stops have categorical voicing in all contexts could be analysed as identical to languages like English [stan1293] and German [stan1295] with aspiration of the FORTIS stops in many contexts and variable presence of voicing in the LENIS series: aspiration in the latter can be either ignored or considered redundant. It was only languages like Icelandic [icel1247], which distinguish between [p^h t^h k^h] and [p t k], that needed a different analysis, since both series are [–voice]: in this case, with a [+spread glottis] specification for the FORTIS stops.

For proponents of Laryngeal Realism, the analysis of such systems should take into account both the phonetic properties of the contrast and the stops’ morphophonological behaviour. Featural specifications are required to be supported by evidence of two kinds. First, the relevant feature’s phonetic correlate (such as voicing or aspiration) should consistently participate in the realisation of the stop series. Second, one should look for (morpho)phonological evidence that the specification is active. For instance, if a stop bears the |voice| specification, then this feature can trigger voicing assimilation, or be restricted from appearing in “weak” positions, such as word- or syllable-finally, resulting in sound patterns such as word-final devoicing. Widespread adoption of versions of Laryngeal Realism has been encouraged by numerous cases where the two sets of criteria – phonetic realisation and phonological activity – point in the same direction. In this way, languages can be classified as belonging to one of a small number of “types”. Thus, English and German have been shown to combine aspiration of FORTIS stops with their phonological visibility on the one hand, and an absence of phonetic or phonological activity of voicing in the LENIS series; conversely, languages like Catalan [stan1289] or Bulgarian [bulg1262] combine categorical voicing of LENIS stops with evidence for phonological activity of this series, while FORTIS stops are both unaspirated and relatively inert phonologically.

This happy confluence has led to the emergence of a particularly strong version of Laryngeal Realism, in which phonetic and phonological criteria must always point in the same direction. Conceptually, this prediction arises because no mechanism within the theory allows for such conflict. Under this view, demonstrating that a particular phonetic cue is obligatorily associated with some category of segments is sufficient to infer the presence of the corresponding featural specification, even in the absence of clear (morpho)phonological evidence (e.g. Beckman et al. 2013). The converse corollary also holds: (morpho)phonological evidence alone can be used to determine a language's place in the typological space. This is particularly useful for historical phonology. While phonetic detail is usually not directly recoverable, one can be on surer ground with respect to morphophonological patterns based on philological or dialectal evidence. If the corollary is correct, one is in a position to make inferences about phonetic detail in the past based on phonological reconstruction. For example, Spaargaren (2009) argues that patterns of laryngeal assimilation in Old English (as in *cēp-te* 'keep.PST.3SG' vs. *freme-de* 'do.PST.3SG', *hīer-de* 'hear.PST.3SG') show greater phonological activity of the [fortis] series of stops, which should lead us to reconstruct Old English [fortis] stops as being aspirated.

This very strong version of Laryngeal Realism has been challenged on a number of grounds. Salmons (2020) argues that a phonetic specification does not necessarily imply the presence of the relevant phonological feature, as in those Germanic languages like Dutch [dutc1256] and perhaps Central Swedish [stan1279] where [lenis] stops are consistently voiced but remain relatively inert phonologically. Kirby & Ladd (2019) question the privileged role of aspiration (voice onset time) in the typology, pointing to the important role of other cues. Most controversially, Cyran (2013) and Iosad (2017) develop case studies to show that phonetic and phonological criteria can be in outright conflict (for a rejoinder, however, see Raimy 2021).

In this chapter, I will use Laryngeal Realism terminology where appropriate to refer to covarying aspects of phonetic and phonological behaviour. With the exception of Breton and possibly several varieties of Scottish Gaelic, the present-day Celtic languages belong to the "aspirating" type, as I will now discuss in more detail.

2.2 The Gaelic languages

In Laryngeal Realism terms, the Gaelic languages clearly belong to the "aspirating" type, with marked [spread glottis] stops in the ⟨p t k⟩ series and unspecified ⟨b d g⟩ stops.

There is near-universal agreement across auditory and instrumental descriptions that the FORTIS stops are postaspirated in both Irish (e.g. Ní Chasaide 1985, Ó Raghallaigh 2014) and the majority of Scottish Gaelic varieties (e.g. Ní Chasaide 1985, Nance & Ó Maolalaigh 2021) in the onset of a stressed syllable. Furthermore, FORTIS stops are preaspirated in postvocalic (or perhaps more precisely postsonorant) position in much of Scottish Gaelic, and likely also at least in some varieties of Irish (Iosad 2025).

Phonologically, |spread glottis| is active in the FORTIS series. The clearest evidence comes from Irish, which has a pattern of coalescence of a LENIS obstruent with a following /h/ that results in a FORTIS consonant. It is observed, for instance, with the future/conditional verbal suffix spelled <-f>: Modern Irish (*Irlig* ‘let.IMP.2SG’ with [gʲ], *lig-f-idh* ‘let-FUT-3SG’ with [kʲ] (Sommerfelt 1960, Ó Buaachalla 1985). Similar behaviour is seen with the passive participle suffix *-tha*: *sníomh* ‘spin.IMP.2SG’ with [v]/[β], *sníofa* ‘spun’.)

Evidence also comes from initial mutation: in lenition, FORTIS stops /pʰ tʰ kʰ/ (and their slender counterparts) alternate with the voiceless fricatives [f h x], and it is widely accepted that such fricatives are phonologically |spread glottis| (“Vaux’s Law”, after Vaux [1998]). Assuming that lenition alters only the manner specification of the stops (*modulo* any necessary place changes), this can be taken as further evidence that the FORTIS stops are |spread glottis|.

The situation in Scottish Gaelic is somewhat less conclusive: due to morphological differences, it does not show as many clear examples of the assimilation pattern.³ It maintains the pattern of mutation, but without clear language-internal evidence for |spread glottis| activity in the fricatives, the argument is much weaker. Nevertheless, there is also no clear evidence against an “aspirating” analysis, and the phonetic data clearly points in the “aspirating” direction just like in Irish.

With regard to the LENIS stops, one is slightly hampered by the lack of detailed instrumental studies, especially of Irish; however, see Ní Chasaide (1985) on Ulster (Gaoth Dobhair). The consensus is that Irish LENIS stops are well-behaved for an “aspirating” language, with variable voicing especially after sonorants (cf. again Ó Raghallaigh 2014). Traditionally, the presence of voicing in the LENIS series is considered an isogloss separating Irish from Scottish Gaelic, which is generally described as having voiceless unaspirated (“short-lag” VOT) LENIS stops (O’Rahilly 1936: 146 sqq.; Ó Murchú 1985, Robinson & Ó Maolalaigh

³The future and conditional are formed without the *-f* element in Scottish Gaelic, and the passive participle is not generally productive. The devoicing pattern can be observed in head-final compounds, but it is not obvious that this is a productive rule, either (cf. Stifter 2015).

2007). This is largely confirmed by instrumental studies such as Ladefoged et al. (1998), although some voicing is possible even here (Nance & Stuart-Smith 2013); Ní Chasaide (1985) also reports (albeit with small sample sizes) some dialectal variation across different parts of the Western Isles.

The literature on Laryngeal Realism does allow for “aspirating” languages to show either variably voiced or fully voiceless LENIS stops. Irish patterns with languages like English and German, whilst Scottish Gaelic is similar to languages like Danish [dani1285] and Icelandic, which are all considered to be “aspirating” by Beckman et al. (2013). Both Irish and Scottish Gaelic also demonstrate a pattern common in Germanic “aspirating” languages whereby the contrast between the two series is neutralized after a tautosyllabic fricative (most notably in the clusters *sp st sk*), with a voiceless unaspirated stop as the outcome. The usual interpretation of such structures (e.g. Salmons 2020) is that the stop and the fricative share a [spread glottis] specification. Thus, from the perspective of Laryngeal Realism both Irish and Scottish Gaelic present as “aspirating” languages.

2.3 The Brythonic languages

The phonetics and phonology of Welsh are likewise straightforwardly compatible with an “aspirating” analysis. The FORTIS stops are postaspirated before a stressed vowel, and at least variably preaspirated after a vowel (Ball & Williams 2001, Jones 1984, Morris & Hejné 2020, Iosad 2023). The LENIS stops are generally described as variably voiced, certainly in South Welsh (Jones 1984); there is a perhaps somewhat surprising dearth of instrumental investigations of North Welsh, but Bell et al. (2023) report a Scottish Gaelic-like system of categorically voiceless LENIS stops. As in the Gaelic languages, the contrast is neutralized to a voiceless unaspirated stop after a tautosyllabic FORTIS fricative.

Phonologically, Welsh is a very clear example of a language showing activity of [spread glottis] in the FORTIS series and inertness of the LENIS stops: for example, as in Irish, a suffix-initial /h/ coalesces with LENIS stops, as in [cy] *gwag* ‘empty’, *gwacáu* ‘to empty’. This evidence is considered in detail by Iosad (2023), to which the reader is referred for further discussion. Thus, Welsh is quite unproblematic as an “aspirating” language.

The phonetic detail of Cornish is not available, but the morphophonological patterns it shows are quite parallel to those seen in Welsh, notably in devoicing (“protection”) patterns that are (residually) attested in the comparative forms of adjectives ([mco] *teg* ‘fair’, *tecka* ‘fair.CMP’, cf. *gwan-ha* ‘weak-CMP’) and in the subjunctive ([d] *jeppr-o* ‘eat.SBJV-3SG’ to *dybry* ‘eat’); cf. Williams (2011).

2.4 Voicing in Breton and Scottish Gaelic

So far, there has been no discussion of any Celtic variety that demonstrates categorically voiced LENIS stops. There are two groups of present-day languages where such representation is found.

The first of these is Breton. As discussed in Iosad (forthcoming), phonetically it is much more reminiscent of a “voicing” language similar to its close neighbour French, with fully voiced LENIS stops and a voiceless unaspirated FORTIS series. However, its morphophonology is more like that of an “aspirating” language. In particular, just like Irish, Welsh, and Cornish, Breton shows a pattern of LENIS stops coalescing with suffix-initial /h/ to their FORTIS counterparts: [br]gleb ‘wet’,⁴ glep-oc’h ‘wet-CMP’ patterns just like W gwlyb, gwlypach. Even patterns superficially resembling those common in “voicing” languages, such as final devoicing, turn out on closer inspection to involve greater markedness of the FORTIS series, meaning that the phonology of Breton is more like that of an “aspirating” language just like its closest relatives. See Iosad (2017) and Iosad (forthcoming).

Consistent closure voicing is also found in some Scottish Gaelic varieties. First, Henderson (1903) reports that all LENIS stops are voiced in “peripheral” dialects, such as Sutherland and South Argyll. Allen (2022) examined the published dialect survey (Ó Dochartaigh 1994) and found consistent voicing in the LENIS series in parts of eastern and south-eastern Sutherland (points 141–147), and variable presence of voicing in north and west Sutherland and Caithness (points 128–140).

Second, closure voicing in stops is associated with the process of post-nasal sandhi (“eclipsis”) across many dialects of Scottish Gaelic (Ó Maolalaigh 1995, Bosch & Scobbie 2009). Both FORTIS and LENIS stops participate in various progressive assimilations following nasals, either word-internally or (a common source of alternations) after nasal-final clitics. There is significant variability in the precise outcomes of these, as Bosch & Scobbie (2009) document, but closure voicing of the stop is one possible result.

The two stop series tend to remain distinct (for instance, as breathy voiced vs. plain voiced stops) in these contexts: Gairloch (Wentworth 2005) *an doras* ‘the door’ [d̪ɔrəs] vs. *an toll* ‘the hole’ [d̪ou̯ˠ] (unmutated [t̪ɔrəs], [t̪ˠou̯ˠ]).⁵ Subse-

⁴The form *gleb* is realised in isolation with a voiceless stop due to final devoicing, but cf. *glebiañ* ‘to wet’.

⁵There is in many, probably most, dialects a strong tendency to show different developments in “close sandhi” (after proclitics and in close compounds like *lon-dubh* ‘blackbird’) and word-internally (as in *cinnteach* ‘certain’): the series tend to remain distinct in the former contexts but often show a tendency towards merger in the latter, with most stops becoming LENIS and realised with voicing. The matter requires much further scrutiny.

quent loss of the nasal may lead to outcomes that can be interpreted in terms of a third distinct phonemic series characterized by a *|voice|* specification; this is proposed, for example, for Applecross by Ternes (2006) and Easter Ross by Watson (2022).

Because of this, we can only rarely analyse postnasal voicing in terms of a “voicing” system: postnasal processes do not generally result in a system contrasting a voiceless (let alone unaspirated) FORTIS series */p^(h) t^(h) k^(h)/* and a consistently voiced LENIS */b d g/*. The only exception here, as Bosch & Scobbie (2009) note, are peripheral dialects of the Sutherland type, but within the broader Celtic context this phenomenon is quite marginal (in more than one sense).

To summarize this section, most present-day Celtic languages are well-behaved representatives of the “aspirating” type under Laryngeal Realism. There is ample phonetic and phonological evidence in favour of associating the FORTIS stops with a featural specification such as *|spread glottis|*, and very little to justify assigning a feature such as *|voice|* to LENIS stops. Even where consistent voicing does occur in the LENIS series, it does not clearly justify a reclassification of the variety in question as representing the “voicing” type. In the next section, the prevailing view of laryngeal contrast in an earlier era is considered.

3 The aspirating view of Proto-Celtic

Unsurprisingly given the situation discussed in Section 2, Proto-Celtic is usually reconstructed with two stop series: a FORTIS category corresponding to “traditional” Proto-Indo-European “voiceless” **p t k kʷ*, with a very early (but not Proto-Celtic) change of **p* to zero, likely via a labial fricative and then **h* in most contexts. The LENIS series **b d g gʷ* represents traditional Indo-European “voiced” and “voiced aspirate” stops. These stops could be either short or long (geminate). While FORTIS geminates were not at all unknown, LENIS geminate stops were rather more marginal, occurring primarily as secondary developments of other clusters or across a morpheme boundary; for the details of gemination in Celtic, see Stifter (2024).

A very simple interpretation is to treat the FORTIS series as (plain) “voiceless” and the LENIS series as “voiced”. However, direct evidence in favour of this account is meagre. Much of the Continental Celtic inscriptional material is not probative at all: it is preserved using writing systems such as the Iberian and Lugano alphabets that generally do not distinguish laryngeal contrast in stops. Celtic material in the Greek and Latin scripts generally uses orthographic *⟨p t c⟩* *resp.* *⟨π τ κ⟩* for the FORTIS stops and *⟨b d g⟩* *resp.* *⟨β δ γ⟩* for their LENIS counterparts. However, there is some potentially useful graphical hesitation. The Gaulish

material in the Latin alphabet presents non-trivial amounts of vacillation: especially frequent is <c> for expected [g], and we also find some confusion between <t> and <d>. Furthermore, a few Greek-script inscriptions witness <θ> and <χ> for the FORTIS stops. Following Gray (1944) and Watkins (1955), this vacillation is often taken to indicate that singleton stops were pronounced differently in Gaulish and in Latin. A common view is that Gaulish /b d g/ possessed some “fortis” property that their Latin counterparts lacked, precluding orthographic <b d g> from representing these LENIS stops: one possibility is that they would not be consistently voiced, being therefore liable to be identified with Latin or Greek voiceless stops.

Given the “aspirating” systems of the present-day languages, it is also possible to simply interpret Proto-Celtic as long-lag VOT (“aspirated”) stops in the FORTIS series. Already, Pedersen (1909) and Lewis & Pedersen (1937) projected the present-day aspiration into Proto-Celtic. In a series of recent publications, Eska (2017; 2018; 2019; 2020) forcefully restated this position within the theoretical framework of Laryngeal Realism, arguing that Proto-Celtic is best analysed as an “aspirating” language, and thus that the present-day systems continue the original pattern unchanged. He adduces evidence of two kinds for this proposition.

In terms of direct evidence, his proposal is supported by the interpretation of the Gaulish material discussed earlier. In addition, Eska (2017) discusses some Celtiberian inscriptions that make a distinction between two stop series,⁶ giving special attention to examples where sequences of a (historically voiceless) fricative and a stop are spelled with the LENIS stop grapheme <sd>. Eska (2017) suggests that this reflects the lack of aspiration after a fricative, much as in present-day Welsh, Irish, and Scottish Gaelic (where similar spellings of such clusters are, or have been, attested in the written tradition).

A second source of evidence is the typology of sound changes affecting Proto-Celtic stops in the descendant languages. For instance, Eska (2018) considers the “loss” of Proto-Indo-European **p* in Celtic: as already noted, this process is normally considered to have proceeded via multiple stages involving voiceless fricatives (**p > *φ > *h > ∅ vel sim.*). Eska (2018) argues that this is compatible with the presence of aspiration on the stop, with reference to Kümmel (2007: 57). Further arguments of this kind are given in Eska (2018) with respect to the Common Celtic period and by Eska (2019; 2020) for the medieval and modern attestations. It is to a re-examination of this typological evidence that I now turn.

⁶The inscriptions in question are analysed by Jordán Cólera (2007); for a recent overview of Palaeohispanic writing systems, see Ferrer & Moncunill (2019).

4 Examining the sound changes

In this section, I examine the typology of sound changes affecting Proto-Celtic stops in the attested languages. After a brief overview of the overall pattern, I will consider the relevant changes by dividing them into four groups: changes that are more likely if Common Celtic belonged to the “voicing” type; changes more compatible with an “aspirating” system; changes where a reasonable, albeit non-trivial case can be made for either possibility; and finally non-probative changes that are easily compatible with either a “voicing” or an “aspirating” pattern in Common Celtic.

4.1 Lenition of stops in Gaelic and Brythonic

Processes of lenition affecting both FORTIS and LENIS stops are a hallmark of Celtic historical phonology. They have been the subject of extensive scholarship over the years, some of which is of foundational importance even outwith the specific concerns of Celtic studies (such as Martinet 1955). In this chapter, however, I am mostly concerned with what these changes reveal about the nature of laryngeal contrast prior to their operation.

The fundamental facts are fairly simple. In intervocalic (or, rather, postvocalic) position, both series of stops in the Insular Celtic languages are reflected with outcomes that are advanced along well-understood lenition trajectories relative to their original state. The developments are exemplified in Table 1. The consonants of interest are bolded, and the transcription of the relevant consonant given when not trivial. As it shows, LENIS stops changed to LENIS fricatives **β ð γ* in both Brythonic and Gaelic languages. In the case of FORTIS stops, there is a difference: in Gaelic, lenition produced FORTIS fricatives, whilst in Brythonic it resulted in LENIS stops. As for geminates, in the present-day languages LENIS geminates **bb dd gg* are (mostly) reflected as singleton LENIS stops, whilst FORTIS geminates again show a split: Brythonic reflects them as FORTIS fricatives **f θ x*, whereas in Gaelic they are reflected as FORTIS stops **p t k*.

Most of these processes are dated to some point in the sub-Roman period of the mid 1st millennium CE (Jackson 1953, Sims-Williams 1990, McCone 1996): they are generally not reflected (even incidentally) in the Continental material⁷ or in the very earliest Romano-British elements borrowed into English. However, they do affect Latin borrowings in both Irish and Brythonic, as well as Brythonic and

⁷But see Section 4.5.1 on the LENIS stops and Stifter (2012) on the possibly related lenition of **s* to **h* in Gaulish.

Table 1: Postvocalic stops in Celtic

Language		Singletons		Gemimates	
		FORTIS	LENIS	FORTIS	LENIS
Proto-Celtic		<i>*katu-</i> ‘battle’	<i>*budaro-</i> ‘deaf’	<i>*(ϕ)likkā-</i> ‘flat stone’	<i>*kred-dī-</i> ‘believe’
Gaelic	Old Irish	<i>cath</i> [θ]	<i>bodar</i> [ð]	<i>lecc</i> [k]	<i>creitid</i> [dʲ]
	Modern Irish	<i>cath</i>	<i>bodhar</i>	<i>leac</i>	<i>creidim</i>
Brythonic	Welsh	<i>cad</i>	<i>byddar</i> [ð]	<i>llech</i> [χ]	<i>credu</i>
	Breton	<i>kad</i>	<i>bouzar</i>	<i>lec’h</i>	<i>krediñ</i>

British Latin borrowings in Irish (McManus 1983), and were certainly complete by the time of the manuscript attestations.

4.2 Evidence for a “voicing” system

This category includes three unrelated changes: early “devoicing” of gemimates, the Brythonic lenition of FORTIS stops to LENIS stops characteristic of Brythonic, and the development of FORTIS stops after nasals in Gaelic.

4.2.1 Geminate devoicing

The development of Common Celtic LENIS geminate stops **bb dd gg* presents difficult problems. They mostly arose from consonant clusters, usually at morpheme boundaries (Stifter 2024): Welsh (W) *credu* ‘believe’, Modern Irish (IR) *creidim* from Proto-Celtic (PCELT) **kred-dī-* (cf. Sanskrit *śraddhā*); W *aber* ‘estuary’ from **ad-bero-*, Old Irish (OIR) *acaldam* ‘colloquy’ (present-day *agallamh*) to the verb *ad:gládathar* ‘address.PRS.3SG’. In general, they are reflected in the present-day languages as short LENIS stops.

That said, there are at least some examples of LENIS gemimates, or LENIS consonant clusters, that have FORTIS outcomes in Brythonic (Pedersen 1909: 159–161; Jørgensen 2022, Stifter 2024: 145–146). Significantly, such examples are mostly root-internal: W *bychan*, Breton (BR) *bihan* ‘small’ (**bikko-*) compared to OIR *bec*, IR *beag* (**biggo-*); BR *bouc’h* ‘blunt’ (**bukko-*) but OIR *boc* ‘soft, tender’, (IR, ScG) *bog* (**buggo-*), and a few others. Also likely related is the reflex of the **sd* (> **zd*?) cluster as θ in cases like W *nyth* ‘nest’, (Middle Breton (MBRET) *nez*, *nyth*, from

**nisdo-* (in contrast to OIr *net*, IR *nead*). The simplest explanation of this development is devoicing of a voiced geminate in Brythonic: as Stifter (2024) points out, it is strongly supported by the *-*tt-* appearing in Gallo-Romance borrowings reflecting Celtic *-*sd-*, such as *pettia* ‘part, piece’ (French *pièce*) to (P)CELT) **k^wesdi-* (OIr *cuit* ‘share, part’, IR, ScG *cuid* but W *peth* ‘thing’, Br *pezh*). The same outcome is visible in Scottish placenames reflecting Pictish **pett-* as *Pet-*, *Pit-*.

This development is not entirely regular, and not all relevant consonants and clusters undergo it. In many morpheme-boundary cases, for instance after the prefix **ek(s)-*, such as W *eglan* ‘seashore’ (**ek-glannā-*), the context is highly vulnerable to cyclic/analogical effects (cf. unprefixated W *glan* ‘shore’), and the data is notably difficult (Russell 1988, Stifter 2024). I am persuaded, however, by Jørgensen’s (2022) argument that devoicing in items like **biggo-* was early, and the irregularities arise because it did not operate after the stabilization of LENIS geminates across historical morpheme boundaries, as in **ad-bero-* ‘estuary’ and **kred-dī-* ‘believe’.

Typologically, devoicing of voiced geminate obstruents and clusters is strongly motivated phonetically (Ohala 1997) and well known typologically: for example, phonetic devoicing of such geminates occurs in Japanese (Hirose & Ashby 2007); and a diachronic/morphophonological pattern of this kind is found in Chaha (Banksira 2000: ch. 2). Under the aspiration-based account, the change from **bb dd gg* to **pp tt kk* must be reconstructed as *[*tt*] > *[*tʰ*] (or even [*ttʰ*]?), with subsequent spirantization. This, I suggest, is unlikely. Degemination of a plain stop to an aspirated one is not an impossible change: indeed, it is commonly cited as a possible source of preaspiration (Blevins & Garrett 1993). The rise of (post)aspirated geminates from plain ones is also attested, for example in Cypriot Greek (Newton 1972, Armostis 2009). However, the ultimate outcome of this change in the Brythonic languages is a fricative: neither preaspirated stops nor geminates of any kind are a plausible immediate source for a fricative.⁸ I conclude that devoicing of voiced obstruent geminates/clusters is a far more preferable interpretation. (On the spirantization seen in Brythonic, see Section 4.4.2.)

4.2.2 Lenition of |fortis| stops in Brythonic

As we saw in Section 4.1, Gaelic and Brythonic differ in how lenition in (roughly) postvocalic position affects singleton FORTIS stops. Here, we focus on their reflexes in Brythonic, returning to the Gaelic pattern in Section 4.4.1. The outcomes

⁸In principle, one could imagine preaspirated stops undergoing preaspiration oralization (e. g. Silverman 2003) with subsequent loss of a stop (**tt* > **h_t* > **θ_t* > *θ*), but this very long sequence of events stretches credulity.

of original FORTIS stops in roughly postvocalic position are LENIS stops: *Wcad* ‘battle’ < **katu-*.

This Brythonic “first lenition” is entirely unproblematic for a “voicing” language: intervocalic stop voicing is both phonetically and typologically trivial. It is more difficult to analyse it as involving loss of aspiration, as would be required under the alternative analysis. Eska (2020) proposes that this change resulted from earlier timing of the glottal spreading gesture in postvocalic position, leading to shorter VOT and reinterpretation of the relevant stops as LENIS. While not impossible, I suggest this explanation is problematic.

Voice onset time – the duration of postaspiration – is clearly an important cue to laryngeal contrast in word-initial position, and the change indeed applies to word-initial consonants in postvocalic sandhi (**esjō teγos* ‘his house’ > Welsh *ei dŷ*). However, it is less clear that VOT is particularly important word-medially. To be sure, there are reports in the literature of languages with consistently postaspirated FORTIS stops word-medially: examples include German (Jessen 1998), Persian (Bijankhan & Nourbakhsh 2009), Northern Icelandic (Pétur Helgason 2002), and Georgian (Vicenik 2010). However, it is not clear what would motivate such a leftward shift in glottal opening after a vowel. Typologically, there are examples of across-the-board timing changes – consonant shifts – but it seems that the more usual case is an increase in VOT rather than a shortening. Grimm’s Law in Germanic and the Armenian consonant shift (e. g. Sayeed & Vaux 2017) are celebrated instances in Indo-European, but we can find parallels elsewhere: for instance, a similar shift has occurred (apparently independently) in at least two branches of Austroasiatic: Angkuic and Mal-Pray (Sidwell 2021). However, it is more difficult to understand such a change as a positionally conditioned pattern with relatively earlier glottal opening specifically in leniting positions.

According to Kümmel (2007: 49), intervocalic lenition of FORTIS stops that is realised as loss of aspiration rather as voicing is something of a northern European specialty: his examples are primarily Germanic varieties such as Icelandic, Danish, and Western Norwegian. I cannot discuss these parallels in detail here, and refer the reader to Iosad (2025), where I suggest that this kind of lenition involves not a shortening of postaspiration, but reduction of preaspiration in FORTIS stops. It is conceivable, but difficult to prove, that a similar scenario applied in Celtic, too. This would give some support to its analysis as an “aspirating” language, although not in the exact way envisaged by Eska (2020).

There is a second possibility. A distribution whereby original FORTIS stops yield aspirated reflexes in “strong” positions, but unaspirated ones (ultimately merging with original LENIS ones) in “weak” positions could result not from

lenition targeting weak positions but from fortition in strong contexts. For example, Hill (2007) argues that Tibetan underwent aspiration of stops in initial position, which yields synchronic distribution analysable as intervocalic deaspiration in Old Tibetan (Hill 2010) or Amdo Tibetan (Green 2012). This possibility cannot be easily rejected for Brythonic. Indeed, it is also consistent with the fact that stops escape (first) lenition not only word-initially but also after a sonorant (W *gwynt* ‘wind’, *serch* ‘love’ with subsequent spirantization): it is widely recognized that the “coda mirror” position {#_, C_} often counts as “strong” (Ségéral & Scheer 2008). However, even if feasible, this scenario presupposes an *absence* of aspiration in FORTIS stops in strong position prior to this change, and thus is not compatible with a view of earlier Celtic as an aspirating language.⁹

To summarize, of the three possible scenarios that have been identified, two imply that Brythonic was a “voicing” language at least prior to the change, and of these one is phonetically trivial; only one is fully compatible with an aspirating system, but relies on rather tentative parallels. On balance, therefore, I suggest that the Brythonic first lenition is much more compatible with a “voicing” system of laryngeal contrast.

4.2.3 Nasal-stop clusters in Gaelic

A third development that strongly supports reconstructing an earlier “voiced” pattern is exclusive to Gaelic, where clusters of a (lenis) nasal followed by a FORTIS stop yield LENIS stops: **kantom* ‘hundred’, (OIr *cét*, Ir *céad*, ScG *ceud*, M *keead* (but W *cant* ‘hundred’). This change has been interpreted in multiple ways, but I suggest that all of them are more compatible with a voicing contrast.

Traditional accounts (Thurneysen 1946, Martinet 1952; see also Stifter 2024) consider the development to have gone through a LENIS geminate stop: **kanto-* > **kēdd-*. This interpretation disaggregates the change into two components, gemination and voicing. I will analyse them separately.

There are two main approaches to the voicing of the stop in the literature. One, endorsed by McCone (1996) under the rubric of “second lenition” (see also Kortlandt 1982), considers the voicing to result from postvocalic lenition that applied to newly postvocalic FORTIS stops after the loss of a preceding consonant: **kant-* > **kēt-* > **kēd-*. This is essentially the same process as the Brythonic “first lenition” (Section 4.2.2), and the same considerations apply to it: it is easy to account for under a “voicing” view, but more difficult, albeit not impossible, under an “aspiration” account.

⁹A similar scenario for the origin of the “aspirating” systems in the present-day Germanic languages is envisaged by Goblirsch (2005), building on Steblin-Kamenskij (1974).

A second view, also discussed by McCone (1996: 91) building on Greene (1960), is that the voicing was triggered by the nasal.¹⁰ This proposition is much more difficult to reconcile with an original “aspirating” system. Nasals can, of course, exert a voicing influence on following stops, but this does not seem to commonly result in a neutralizing loss of aspiration. In Section 2.4, I discussed the postnasal voicing processes of Scottish Gaelic: generally, the outcomes of nasal + FORTIS stop sequences are distinct from underlying LENIS stops, and when they do result in a merger, it is precisely where the FORTIS stops show little if any aspiration.

A close parallel is provided by (Amur) Nivkh [gily1242]. It contrasts aspirated FORTIS stops [p^h t^h c^h k^h q^h] with voiceless unaspirated LENIS stops [p t c k q], and possesses a postnasal voicing process quite similar to the Scottish Gaelic “eclipsis”. Crucially for our purposes, it affects LENIS (unaspirated) stops, causing them to become voiced (*təf-ku* ‘house-PL’ but *qan-gu* ‘dog-PL’, but fails to affect FORTIS (aspirated) stops (*alɣ-p^he* ‘pick berries’ but *t^hobzan-p^he* ‘pick nuts’).¹¹ More generally, Kümmel (2007: 54) finds that postnasal voicing of aspirated stops usually produces aspirated or breathy-voiced outcomes. If anything, postnasal position can be associated with plain stops becoming aspirated, as attested in numerous Bantu languages (Downing & Hamann 2018).¹² I conclude that a **nt > *(n)d* scenario is much more strongly compatible with a view where the contrast is of the “voicing” type.

The second potential component of the change, gemination, is perhaps less relevant for our concerns (see Section 4.5.2 below). The literature widely accepts it to have occurred in these clusters, even though the evidence is rather meagre. If gemination did occur, it could be reconstructed as assimilation of **nd* clusters produced by postnasal voicing (this finds a good parallel, for instance, in a number of Semitic languages), or a direct **nt > *dd* change, with a parallel in the Western Sámi languages (Sammallahti 1998; see further Kümmel 2007: 180–181). As far as can be seen, the outcomes of this process merge with original degeminated LENIS stops, as in OIr *bec* ‘small’ (**biggo-*), but the dating of the degemination is unclear: as long as it follows the (likely very early) spirantization of original **b d g* (see Section 4.5.1 below), it could be dated to almost any time in

¹⁰Such clusters do not merge with original **nd*-type sequences, because in the latter the nasal became FORTIS and was preserved: **angu- > *eNgu- > ing* ‘straight’, in contrast to **anku- > *engu- > éc* ‘death’.

¹¹Examples from Gruzdeva (2024: 502).

¹²Hyman (2001) notes a synchronic postnasal deaspiration process in the Nguni languages [ngun1276], but diachronically it represents the blocking of aspiration of plain voiceless stops after nasals. If anything, this pattern *supports* the proposition that FORTIS stops were originally plain.

the history of the Gaelic languages. Whether or not the original nasal-stop clusters went through a geminate stage does not influence my conclusions as to the nature of laryngeal contrast.

4.3 Evidence for an “aspirating” system

Two sound changes provide strong support for an “aspirating” system of laryngeal contrast. They are both late in the relative chronology, because they specifically affect “secondary” clusters that arose via a process of unstressed vowel syncope. This syncope can be dated to the early to mid 6th century CE (Jackson 1953, Sims-Williams 1990).

4.3.1 Voicelessness assimilation

In clusters created by syncope where the second member was a FORTIS obstruent (not least [h]), one can observe regressive laryngeal assimilation. These processes occur in both Brythonic and Gaelic. For Brythonic examples, see W *ateb* ‘answer’ from **ad-heb-* < **ati-sek*^w-, *drycin* ‘storm’ < **drug-hin-* < **druko-sin-*. In Gaelic, there are examples like OIR *dephthigim* ‘fight.PRS.1SG’, IR *deafach* ‘contentious’ (OIR *debthach*) by contrast with *debuith* ‘discord’, IR *deabhaidh* (McCone 1996: 128). For the behaviour of *h* in Gaelic, cf. *imb:soí* ‘turn.PRS.3SG’ but prototonic *-impai* from **ambi-sowet(i)*. As I discussed in Section 2.1, this kind of pattern, which privileges FORTIS phonological activity, is very strongly compatible with (if not diagnostic for) an aspirating system.

4.3.2 Provection

A second post-syncope development, relevant only for Brythonic, also concerns secondary obstruent clusters: LENIS geminates arising from syncope yield FORTIS stops in the present-day languages, as in Welsh *llety* ‘lodging’ from **led-dey*-.¹³ Under an account like Eska’s (2020), in which the LENIS stops were voiceless, the development, commonly known as “provection”, can be interpreted as a change from a voiceless geminate to a (preaspirated) stop, which I already noted to be typologically well supported. Under an account based on voicing, the development needs to proceed via an intermediate stage of geminate devoicing: this is, as already noted, unproblematic phonetically, but makes the derivation more complex. Overall, this process does not appear to be fully probative, but an aspirating analysis is more parsimonious in this case.

¹³Cf. also the placenames Lichfield (Staffordshire) and Lytchett (Dorset) from Brythonic **léd-gēd-* < **lēto-kaito-* ‘grey forest’ (Jackson 1953: 563).

4.4 Ambiguous changes: Spirantization of FORTIS stops

One change that has a rather ambiguous status is the spirantization of FORTIS $^{*}(p) t k$ stops to the FORTIS non-strident fricatives $^{*}f \theta x$. It occurs separately, with different conditioning, in the two branches.

4.4.1 Goidelic first lenition

As discussed in Section 4.1, FORTIS fricatives $^{*}\theta x x^w$ are the usual outcome of original singleton FORTIS stops in Goidelic in postvocalic position (including in postvocalic sandhi). The usual explanation is that the stops were originally aspirated and became fricatives likely via an affricate stage, a change with multiple typological parallels (not least the High German Consonant Shift); see Koch (1990), Isaac (2004). Eska (2018) further assimilates the spirantization of Proto-Indo-European $^{*}p$ to $^{*}\phi$ in Early Celtic to this parallel, stretching the possible presence of aspiration even earlier.

In support of this proposition, Eska (2018) cites the findings of Kümmel (2007: 57). However, these results warrant further scrutiny: what Kümmel (2007) finds is that aspiration is necessary for FORTIS spirantization when it is an across-the-board consonant shift, but both changes of individual consonants (as with PCelt $\phi < \text{PIE } ^{*}p$) and positionally restricted lenition processes are attested as spirantizations of what are unaspirated FORTIS stops. One particularly well-known example of the latter is *gorgia toscana*, the postvocalic lenition of singleton FORTIS stops found in Tuscan Italian [ital1282]. In a recent in-depth discussion of this process, Marotta (2008) argues that it does not proceed via affrication of aspirated stops, although see Honeybone (2023) for some skepticism on this possibility. Another possible case is reported from Southern Jutland Danish by Puggaard-Rode (2024).

My conclusion is that the spirantization of FORTIS stops cannot clearly distinguish between the two approaches to laryngeal contrast. It is more easily compatible with an “aspirating” system, but a “voicing” pattern cannot be conclusively ruled out at this stage.

4.4.2 Brythonic second lenition

In Brythonic, FORTIS stops have fricative outcomes in two contexts. First, this is observed regularly with postvocalic FORTIS geminates: W *llech* ‘slate’, Br *lec’h*, Cornish *lehan* $< ^{*}(\phi)likk\bar{a}$. Second, the same outcome is found for FORTIS singletons in two contexts: after a liquid (*serch* $< ^{*}sterk\bar{a}$ ‘love’; contrast OIr *sercc*, Ir *searc*) and in sandhi after the loss of certain word-final consonants, especially $^{*}h$

from earlier **s* (*esjas teyos* ‘her house’ > **ehja θeyoh* > W *ei thÿ*; contrast OIR *a (t)tech*, IR *a teach*).

A long line of work (Jackson 1953, 1960, Koch 1989, 1990, Isaac 2004, 2008) has taken this development to support the presence of aspiration on FORTIS stops: the greater “force” of geminate stops is hypothesized to have facilitated aspiration, followed by affrication and subsequent spirantization (**tt^h* > **t(t)θ* > *θ*).

However, authors such as Thomas (1990), Sims-Williams (1990, 2008) and McCone (1996) have suggested this is unnecessary: instead, they analyse the Brythonic developments as a combination of FORTIS degemination (non-probative for our purposes, Section 4.5.2) followed by spirantization of the type discussed in Section 4.4.1 (so **likkā*- > **likā*- > *llech*). I find the latter scenario more persuasive. In particular, the direct development from geminates to fricatives without degemination is typologically highly suspect (e. g. Kirchner 1998). This analysis also allows us to dispense with the otherwise highly odd gemination that the traditional scenario has to posit to account for spirantization of singletons in forms like W *serch*, *ei thÿ*. Instead, all of these developments can be subsumed under a “second lenition”, where postvocalic singleton FORTIS stops of secondary origin (i. e. produced by degemination or by loss of a preceding consonant), as well as post-liquid stops, are reflected as FORTIS fricatives, parallel to the “first lenition” in Gaelic. If one accepts the degemination-and-lenition scenario, the Brythonic development is now a combination of two non-probative changes.

4.5 Non-probative changes

Finally, I briefly consider two categories of changes that are not probative for the nature of the Early Celtic system, being easily compatible with either option.

4.5.1 Spirantization of |lenis| stops

Original LENIS singleton stops yield LENIS fricatives, generally reconstructed as **β ð γ*, in both Gaelic and Brythonic. They are mostly reflected as fricatives, usually (but sometimes variably) voiced, in the present day (Ball & Williams 2001). Intervocally, they often undergo further lenition processes such as debuccalization and ultimately deletion, especially in Irish and Breton, but the general pathway is widely agreed on.

Although traditionally this process is dated to the same mid 1st millennium CE stage as the other Insular lenitions, more recent scholarship (e.g. Sims-Williams 1990; Villar 1993; McCone 1996; Eska 2017) has been willing to date it to a much

earlier period. The reason is that there are some traces of this process (such as loss of intervocalic *g and fricative spellings of postvocalic *d) in both Gaulish and Celtiberian.

This change is not especially informative. Voiced fricatives are commonly produced by lenition of voiced stops, but they can also arise from the spirantization of voiceless, contrastively unaspirated stops. *Comparanda* for the lenition of voiced stops to voiced fricatives are trivial to find: for instance, this process is pervasive in Romance languages, including Spanish [stan1288], Catalan, and many languages of Italy. However, clearly “aspirating” languages can also show a pattern of LENIS voiceless unaspirated stops alternating with voiced continuants. I have already cited Nivkh. The two series of stops, both aspirated /p^h t^h c^h k^h q^h/ and voiceless unaspirated /p t c k q/, participate in a morphologically restricted process of morpheme-initial consonant mutation. They alternate with voiceless and voiced continuants [f ɾ s x χ] and [v r z ɣ ʁ] respectively, just like in Scottish Gaelic: *təf-ku* ‘house-PL’ but *mu-yu* ‘boat-PL’. I conclude that the lenition of singleton LENIS stops to voiced fricatives, whether treated as part of the Insular lenition processes or put further back in time, does not furnish probative arguments regarding the nature of laryngeal contrast at the time that it occurred.

4.5.2 Degemination

The clearest case of degemination is found with FORTIS geminates in the Gaelic languages, where they are reflected as FORTIS stops (with, where appropriate, preaspiration): **(ϕ)likkā-* ‘flat stone’ > IR, ScG, *leac*, M *lhiack*. There is no contrastive consonant (or in any case stop) length in the present-day languages, so these stops are generally considered to be short (with a few exceptions). This implies a degemination at some point along the way. Unfortunately, there is no consensus as to when precisely it occurred: the possibilities range from the date prior to the manuscript record to, in some cases, incomplete degemination up to the present day (on issues of consonant quantity in Gaelic, see recently Lewin 2023, Stifter 2024, Wheatley & Iosad in preparation). Whatever the case may be, degemination itself is a trivial change that does not offer a criterion to distinguish between the possible systems of laryngeal contrast.

5 Reconsidering the evidence

Having evaluated relevant changes individually, I can now synthesize the evidence for the two branches. In both cases, I will argue that the earlier changes

are overall more compatible with a voicing-based system, while syncope marks the advent of the present-day “aspirating” patterns.

5.1 Gaelic

Table 2 summarizes the developments in the Gaelic languages. To facilitate interpretation, cells corresponding to strongly diagnostic changes are shaded. The degemination of FORTIS stops is difficult to place in the chronology, and it is in any case not probative, so it is excluded. As for LENIS geminates, one could perhaps place the degemination of the original ones relatively early, but since any geminates that arose due to syncope appear to have been shortened on par with the earlier ones, for the sake of concreteness I show a single post-syncope round of degemination (in agreement with Stifter 2017: 1199).

Table 2: Consonant developments in Gaelic

Early changes	Voicing scenario	Aspiration scenario
(p) t k > (f) θ x / _V <i>*katu-</i> > <i>cath</i>	Compatible	Strongly compatible
(p) t k > (b) d g / _N <i>*kant-</i> > <i>cét</i>	Strongly compatible	Weakly or not compatible
Post-syncope changes		
bb dd gg > b d g <i>*biggo-</i> > <i>bec</i> <i>*ag-gald-</i> > <i>acaldam</i>	Not probative	Not probative
Assimilation <i>*-ámbi-sowet(i)</i> > <i>-impai</i>	Unlikely	Strongly compatible

My interpretation of Table 2 is that the earlier sound changes are very strongly compatible with a “voicing” system of laryngeal contrast, whereas an “aspirating” system is only very weakly, if it all, compatible with the data. The behaviour of nasal-stop clusters in particular is very difficult to square with an “aspirating” analysis. Admittedly, the evidence for spirantization of unaspirated voiceless stops is problematic in some ways, but the phenomenon appears to be not entirely unattested.

On the other hand, the later, post-syncope pattern is essentially what is found in the present-day languages, which show prototypical “aspirating” behaviour.

Therefore, the prudent assumption is that secure evidence for the “aspirating” system does not date to earlier than around the time of syncope, late in the Primitive Irish period, rather than to an earlier period in the history of Celtic, *pace* Eska (2018).

I suggest there is some circumstantial evidence that supports reconstructing the rise of aspiration in Gaelic FORTIS stops to the late Primitive Irish period. Almost all present-day Gaelic languages show a process of vowel intrusion in heterorganic sequences of a sonorant and another consonant (as in Irish [dʲarəɡ] *dearg* ‘red’, OIR *derc*); see Morrison (2019), Iosad & Maguire (in preparation) for details. Today, this process is regularly blocked when the second consonant in such a cluster is a FORTIS stop: such clusters lack intrusive vowels and in many dialects, especially in Scottish Gaelic, voiceless sonorants are found instead ([ɔ̟k] *olc* ‘evil’). Morrison (2019) persuasively argues that the intrusive vowel arises whenever the sonorant is moraic in the stem-level phonology. If this is correct, the FORTIS stop exception can be understood as failure of the sonorant to acquire a mora before such stops. Morrison (2019) points to East Perthshire Gaelic (Ó Murchú 1989), where the synchronic generalization is that coda sonorants are long (thus moraic) before historically LENIS, but not historically FORTIS stops ([pal:k] *balg* ‘bag’ but [ɔ̟k] *olc* ‘evil’), a pattern Morrison (2019) considers archaic. One can hypothesize that sonorants before FORTIS stops fail to acquire a mora precisely because they were devoiced by the following stop (on the connection between voicing and moraicity, see Zec 1988), a phenomenon closely allied to preaspiration.

Now, although the exceptions to the generalization that FORTIS stops block vowel intrusion are marginal in the present-day languages, Eska (2010) identified a number of Ogam inscriptions that witness what he calls “non-canonical” vowel intrusion, with examples such as ⟨ERACİAŠ⟩ (cf. ⟨ERCIAS⟩ and later *Ercc*). He argues, convincingly, that, *pace* O’Rahilly (1936), Hind (1996), such spellings do reflect the presence of intrusive vowels in these contexts in Primitive Irish. If he is correct, then one can posit that intrusion was able to apply before FORTIS stops precisely because such stops did not trigger sonorant devoicing. Once devoicing set in, it blocked sonorant moraicity, and the intrusive vowels were suppressed. Given that we observe the intrusive vowels in this context in the Ogam corpus, but not in the later Gaelic languages, I suggest that the rise of sonorant devoicing, associated with aspiration, may be tentatively dated to the late Primitive Irish period – exactly as suggested by the analysis of Table 2.

5.2 Brythonic

Turning now to Brythonic, Table 3 shows the sequence of changes and their evaluations. I place the degemination of stops early in the chronology, because in the case of FORTIS stops it feeds second lenition, whilst LENIS stops degeminated prior to syncope, since they escape the devoicing experienced by secondary LENIS clusters.

Table 3: Consonant developments in Brythonic

Early changes	Voicing scenario	Aspiration scenario
p t k > b d g / _V <i>*katu-</i> > <i>cad</i>	Strongly compatible	Weakly compatible
Sporadic geminate devoicing <i>*bigg-</i> > <i>bikk-</i>	Strongly compatible	Highly unlikely
Degemination <i>*bikk-an-</i> > <i>*bikan-</i> > <i>bychan</i> <i>*likkā-</i> > <i>*likā-</i> > <i>llech</i> <i>*ad-bero-</i> > <i>*abbero-</i> > <i>aber</i>	Not probative	Not probative
p t k > f θ x / _V <i>*broko-</i> > <i>broch</i> <i>*sterkā</i> > <i>serch</i>	Compatible	Strongly compatible
Post-syncope changes		
bb dd gg > p t k <i>*leto-teyos</i> > <i>*led-dey-</i> > <i>llety</i>	Strongly compatible but more complex	Strongly compatible
Assimilation <i>*ati-sek^w-</i> > <i>*ad-heb</i> > <i>ateb</i>	Unlikely	Strongly compatible

As in Gaelic, later developments, like the present-day patterns, agree strongly with the “aspirating” analysis. For the earlier stages, it would appear that both analyses are feasible. The balance might be tilted in favour of the voicing approach (as for Gaelic) if one considers that lenition as deaspiration (or preaspiration shortening) is somewhat rare typologically, and accepts the probative value of sporadic degemination despite its lack of regularity.

It is telling that the very earliest changes in Brythonic are also the ones most strongly pointing towards a “voicing” analysis. Just as with Gaelic, the closer

one gets to the present day, the more the balance tips towards “aspirating” patterns, even though (with the problematic exception of geminate devoicing) a very strongly diagnostic pattern like the postnasal voicing of Gaelic cannot be found.

Overall, I conclude that while it is *possible* that FORTIS stops were aspirated at a very early stage of Celtic, as argued by Eska (2017; 2018; 2019; 2020), the evidence from the earliest sound changes is also compatible with a view of FORTIS stops as voiceless (likely unaspirated) and LENIS stops as voiced. The change to the present-day system can be tentatively dated to some point late in the era of considerable upheaval in the structure of the Insular Celtic languages in the mid 1st millennium CE.

6 Conclusions

Overall, I have argued that while most modern languages (other than Breton) are classic “aspirating” systems in the sense posited by Laryngeal Realism (in agreement with Eska 2018), solid evidence for these patterns does not stretch much further back than the sub-Roman period in Britain and Ireland, when the Celtic languages underwent a series of drastic phonological and morphological changes that have largely defined their shape in the medieval and modern period. In this final section, I offer some possible perspectives on the wider relevance of this conclusion for Celtic historical linguistics.

In particular, I suggest that the proposed interpretation does not necessarily mean that Early Celtic was necessarily a “voicing” language *tout court*. Rather, I suggest that we should take seriously the proposition that the Celtic languages, agreed to have been spoken over a large part of Western Europe throughout the 1st millennium BCE, demonstrated a degree of variability in their laryngeal contrast. Although the general slant of the reconstruction proposed here is that the development went from a “voicing” towards an “aspirating” system, the change need not have happened only once and/or in the same place.

Instead, I suggest that it is wise to envisage a pool of synchronic variation (Ohala 1989) in the realisation of laryngeal contrasts across prehistorical Celtic varieties, with different dialects undergoing often similar shifts at different points in time. Such a régime would explain, for example, the evidence discussed in Section 3, which seems to suggest an “aspirating” system being attested on the Continent several centuries before my proposed change in the Insular varieties. If “aspirating” patterns did indeed stabilize earlier in Continental Celtic, it may not be a coincidence that it was precisely the Continental varieties that were located near the spatial interface with Germanic (for up-to-date discussion of

Celtic-Germanic relationships, see van Sluis et al. 2023), widely accepted to have been just such an “aspirating” language from a very early stage (Salmons 2020).

Even within Insular Celtic, it is not clear that the change happened at the same time. Even if a very late date for the split of Goidelic from Brythonic is accepted (perhaps along the lines of Schrijver 2015), it would have already occurred by the time of syncope, a likely *terminus post quem* at least for the Gaelic languages. The shift towards an “aspirating” system may thus have been at least partly independent in the two Insular groups. In this way, this change presents another example of Sapirean drift driven by the narrowing down of the range of inherited variation (Joseph 2006, Iosad 2025).

A second corollary is that if some version of a “voicing” pattern was maintained at least in some part of the Celtic *Sprachraum* as late as the first millennium CE, then its occurrence in Breton does not necessarily have to be fully ascribed to the influence of Romance. If some aspects of a “voicing” system were maintained during the spread from Britain, then perhaps the role of contact could have been more complex, which may have consequences for our understanding of the sociolinguistics of Romance-Celtic interaction in Brittany.

More widely, I suggest that a model that takes synchronic variation in proto-languages seriously (cf. Natvig & Salmons 2020) is potentially productive in tackling many classic issues in Celtic historical linguistics. I leave additional elaboration of such a model to further research.

Abbreviations

BR	Breton	PCelt	Proto-Celtic
IR	Modern Irish	PIE	Proto-Indo-European
M	Manx	ScG	Scottish Gaelic
MBRET	Middle Breton	SGDS	Survey of the Gaelic Dialects of Scotland
MCo	Middle Cornish		
OIR	Old Irish	W	Welsh

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